# Mohamed Brahimi

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## Education

**University of Toronto** Sept. 2025 – Expected May 2029

*BSc. Statistics Co-op, Minors in Computer Science, Geographic Information Systems* *Toronto ON, CA*

- Relevant Courses: Intro to CS II (data structures, graph algorithms in C; OOP in C++), GIS Principles and Applied Mapping (mapping, spatial analysis, PostgreSQL), Intro to Probability, Calculus II

## Experience

**Education & Web Coordinator** Oct. 2025 – Present

*UTSC Sports Business Association* *Toronto, ON, CA*

- Built and deployed the club website in Next.js, React, and Tailwind CSS, scoring 100 (desktop) and 94 (mobile) Performance and 100 Accessibility, Best Practices, and SEO on Lighthouse, with image optimization and lazy loading.

**Web & IT Developer** Jun. 2026 – Aug. 2026

*Muslim Welfare Canada* *Toronto, ON, CA*

- Deployed and documented a centralized NAS server utilizing RAID 5 for hardware fault tolerance and data redundancy, implementing access controls to secure sensitive files for 100 users across 3 office locations
- Redesigned multiple website pages, and added a fundraising page which raised close to $50,000 for a charity event
- Collaborated with front-end developers on a React-based website overhaul

## Projects

**Atlas** | *React, TypeScript, Mapbox GL JS, GeoJSON, Vite, CSS* Oct. 2026

- Contributed the interactive map to a Gemini-powered news investigation platform built in 48 hours by a team of 4 at Hack the Valley 11
- Built the core Mapbox GL JS map component in React/TypeScript which renders locations through 7 layers, with point clustering and click-to-expand
- Rendered city/region records as boundaries and fixed a bug by splitting antimeridian bounds into MultiPolygons
- Wired the map to backend endpoints and Gemini-driven actions, flying the camera to locations by validated ID rather than model-generated coordinates

**TOQuery** | *Python, FastAPI, PostgreSQL, PostGIS, Docker, GitHub Actions* Oct. 2026 - Present

- Architected a FastAPI + PostGIS REST API serving Toronto data as GeoJSON, with automated tests running on every push through a GitHub Actions CI pipeline
- Wrote a GeoPandas ingestion pipeline that deduplicates records, repairs invalid geometries, validates the CRS, and batch-loads into PostGIS with row-count verification
- Containerizing the API and database with Docker Compose and built a Leaflet demo map that renders layers live

**FBRef Stats** | *Python, Requests, Pandas, Beautiful Soup 4* Jan. 2025

- Created and published a Python library on PyPI for scraping FBRef soccer data, with 16,875+ downloads
- Built a 12-method API across 10 leagues returning league, squad, and scouting data as Pandas DataFrames
- Parsed leaderboard tables embedded in commented-out HTML, added request throttling to keep scraping reliable, and documented with a README and usage examples

**Cycletrace** | *Linux, Python, GeoPandas, Supabase, Vite, React, Leaflet* Aug. 2026 - Present

- Built a Raspberry Pi GPS tracker in Python logging real-time location, speed, distance, and trip data
- Designed an offline-first sync pipeline: rides are logged to SQLite with a synced flag and uploaded to Supabase in batches, so interrupted syncs resume where they stopped
- Developed a Vite/React dashboard that renders synced routes from Supabase and computes trip statistics (distance, average/max speed, elevation)

## Technical Skills

**Languages**: Python, TypeScript, JavaScript, SQL, HTML/CSS, C & C++ (coursework)
**Frameworks & Libraries**: React, Next.js, Vite, Tailwind CSS, FastAPI, Mapbox GL JS, Pandas, GeoPandas, Requests, BS4, Leaflet
**Databases**: PostgreSQL, PostGIS, SQLite, Supabase
**Dev & GIS Tools**: Git, GitHub Actions, Docker, Linux, Raspberry Pi, Arduino, QGIS, ArcGIS Pro